

A PROJECT REPORT ON
“ UNITY GAIN BUFFER USING TWO STAGE
OPAMP ”

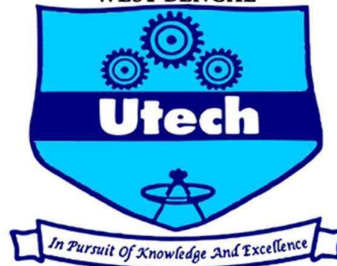
Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor Of Technology in Electronics Engineering

(VLSI Design and Technology)

from

MAULANA ABUL KALAM AZAD
UNIVERSITY OF TECHNOLOGY,
WEST BENGAL



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I wish the candidates the very best for their endeavors.

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Abstract

This project presents the design, implementation, and performance evaluation of a two-stage CMOS operational amplifier configured as a unity gain buffer using the Cadence Virtuoso design environment. The primary objective of the design is to achieve stable voltage buffering with high input impedance, low output impedance, and near-unity voltage gain, making the circuit suitable for analog signal conditioning and impedance matching applications.

The operational amplifier was designed using CMOS technology and analyzed through DC, AC, and transient simulations. The AC analysis was performed to evaluate the frequency response, stability, and bandwidth of the amplifier, while transient analysis was used to examine its large-signal performance. The simulation results demonstrated a unity-gain bandwidth of 3.2471 MHz and a phase margin of 66.64° , confirming stable operation under unity-feedback conditions. Based on the design parameters of a $15\text{ }\mu\text{A}$ bias current and a 0.2 pF load capacitance, the theoretical slew rate was calculated to be $75\text{ V}/\mu\text{s}$, indicating good dynamic response to rapidly varying input signals.

The obtained results verify that the designed unity gain buffer provides reliable voltage tracking, excellent stability, and satisfactory frequency and transient performance. The circuit is well suited for applications such as impedance matching, signal buffering, sensor interfaces, data acquisition systems, and mixed-signal analog front-end circuits. This project demonstrates the effectiveness of CMOS operational amplifier design and Cadence simulation techniques in developing high-performance analog integrated circuits.

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INTRODUCTION

Operational amplifiers (op-amps) are one of the most widely used building blocks in analog electronic circuits. They are used in a variety of applications such as signal amplification, filtering, voltage comparison, data acquisition, and sensor interfacing. Due to their versatility and high performance, operational amplifiers are an essential component in modern analog and mixed-signal integrated circuits.

A unity gain buffer, also known as a voltage follower, is one of the simplest and most important configurations of an operational amplifier. In this configuration, the output is directly connected to the inverting input, while the input signal is applied to the non-inverting terminal. The circuit provides a voltage gain of approximately one, meaning the output voltage follows the input voltage. Although it does not amplify the signal, it offers high input impedance and low output impedance, making it useful for impedance matching and preventing loading effects between different stages of a circuit.

In CMOS analog design, operational amplifiers are commonly implemented using a two-stage architecture, which provides sufficient voltage gain and output swing while allowing stable operation with proper compensation. The design of such amplifiers involves selecting appropriate transistor dimensions, establishing suitable bias currents using current mirrors, and analyzing important performance parameters through simulation.

The objective of this project is to design and simulate a two-stage CMOS operational amplifier configured as a unity gain buffer using the Cadence Virtuoso design environment. The circuit is designed using CMOS technology and its operation is verified through DC, AC, and transient analyses. The project also aims to study the design methodology of CMOS operational amplifiers and understand the factors that influence their performance, stability, and overall operation.

TABLE OF FIGURES

Element Name	Number of Elements
P-channel MOSFET	3
N-channel MOSFET	5
Voltage Source	
• AC Source	1
• DC Source	2
Current Source	1
Capacitor	2

OBJECTIVES

1. Design and Implementation

To design a two-stage operational amplifier configured as a unity gain buffer, ensuring high input impedance and low output impedance for effective signal isolation.

2. Performance Optimization

To achieve stable operation with minimal phase shift, wide bandwidth, and low distortion while maintaining unity voltage gain.

3. Practical Application

To validate the buffer's ability to interface between high-impedance signal sources and low-impedance loads without signal attenuation. The use of a two-stage op-amp in this design offers several advantages, including enhanced drive capability, better common-mode rejection ratio (CMRR), and improved slew rate performance compared to a single-stage design. This makes the buffer suitable for applications such as sensor signal conditioning, analog-to-digital converter (ADC) interfacing, and audio signal processing.

4. Simulation and Verification

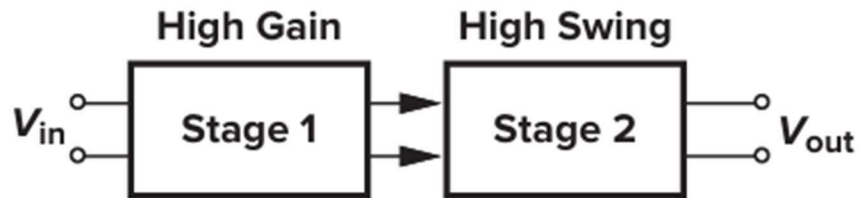
To simulate the circuit using EDA tools (e.g., Cadence Virtuoso) and verify performance parameters such as gain, bandwidth, slew rate, and stability.

5. Analysis and Documentation

To analyze the trade-offs between gain, stability, and power consumption, and document the design process for academic and practical reference.

ARCHITECTURE

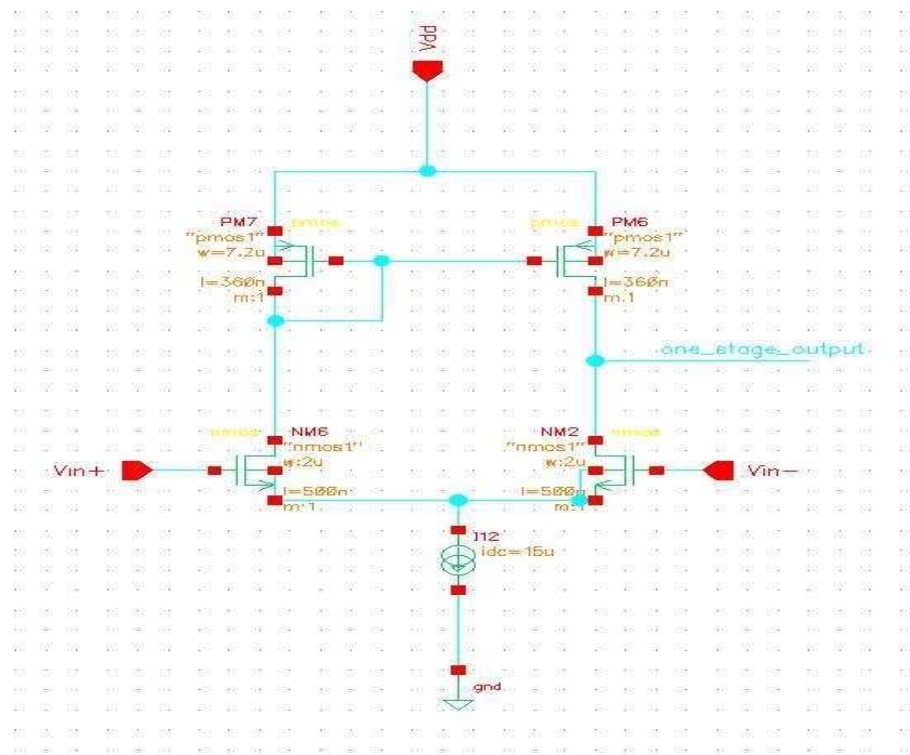
The designed circuit is based on a two-stage CMOS operational amplifier configured as a unity gain buffer (voltage follower). The architecture consists of four main functional blocks: a differential input stage, a current mirror, a second gain stage, and a compensation capacitor.



Two stage op-amp Architecture

1. DIFFERENTIAL INPUT STAGE:

The first stage of the operational amplifier consists of an NMOS differential pair with a PMOS active load. This configuration provides



differential-to-single-ended conversion while achieving high voltage gain and improved transconductance efficiency. The input stage is responsible

for amplifying the input voltage difference and contributes significantly to the overall noise performance of the amplifier. The selection of the input transistor type and bias current is optimized to achieve a balance between input-referred noise and power consumption.

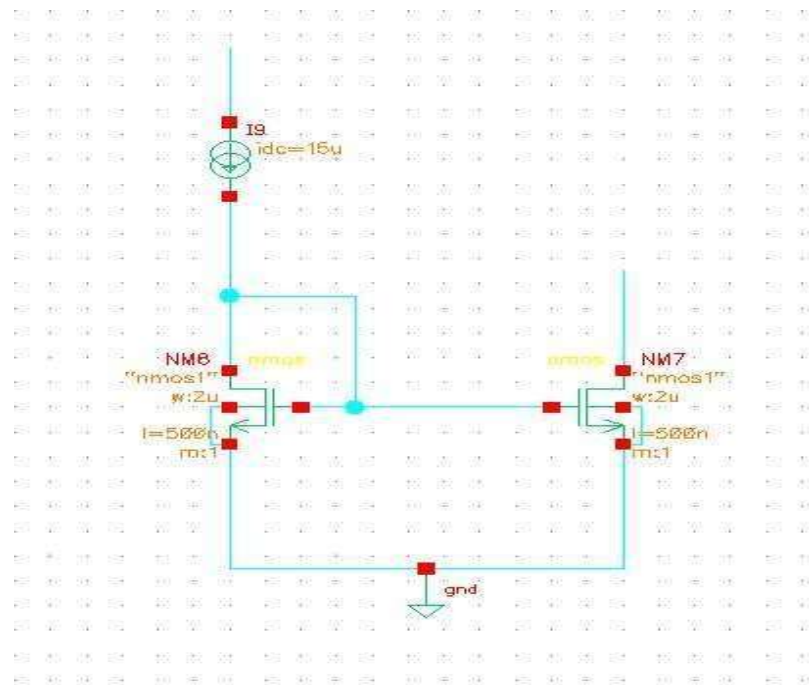
2. CURRENT MIRROR

In a MOS current mirror, one transistor is diode-connected to establish a reference current. This current sets the gate-source voltage V_{GS} , which is applied to the gate of the second matched transistor. As both transistors have the same V_{GS} , they conduct nearly equal drain currents, thereby replicating the reference current at the output.

Use in the Circuit

In the designed two-stage CMOS operational amplifier, the current mirror is used to:

- Generate stable bias currents for different stages of the amplifier.

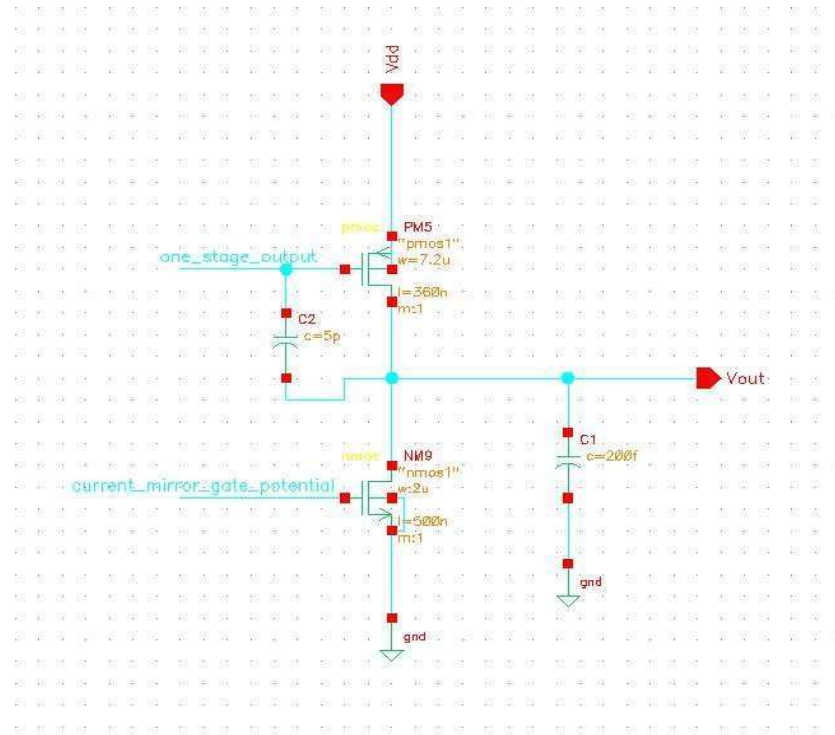


- Act as an active load for the differential amplifier, thereby increasing voltage gain.
- Improve current matching and reduce the need for large resistor networks.
- Enhance circuit stability while reducing power consumption and silicon area.

Thus, the current mirror is an essential building block that ensures proper

biasing and reliable operation of the operational amplifier.

3. CMOS AMPLIFIER



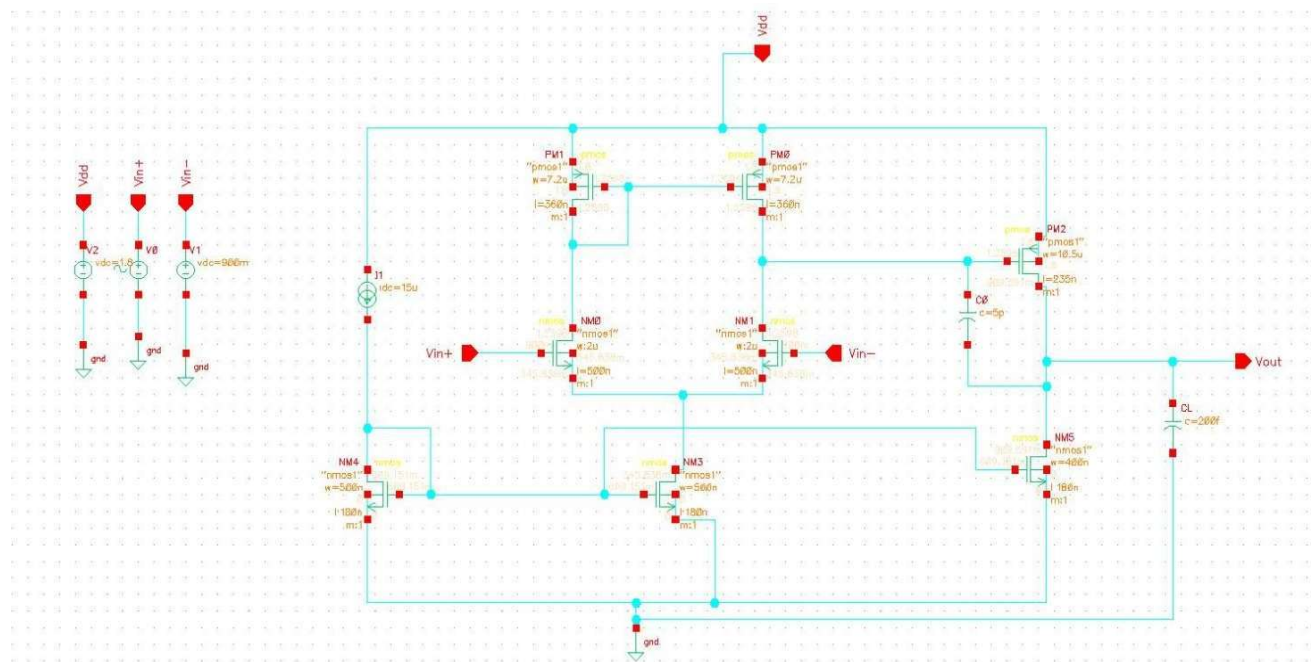
The output of the differential stage is fed to a common-source amplifier, which forms the second stage. This stage provides additional voltage gain and supplies sufficient output drive capability for the load.

4. COMPENSATION CAPACITOR

A Miller compensation capacitor is connected between the first and second stages to improve the frequency response and ensure stable operation under negative feedback. It introduces a dominant pole that increases the phase margin and prevents oscillations.

5. OPEN LOOP GAIN

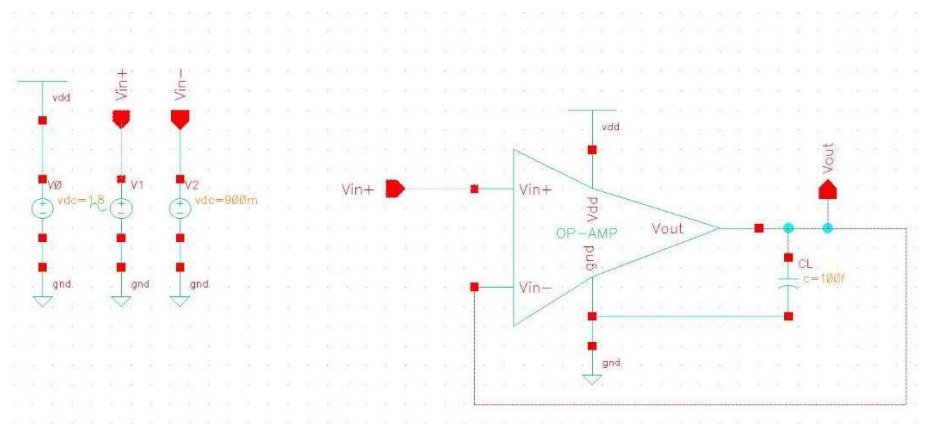
- The output is taken from the second gain stage.
- In open-loop configuration, the output voltage is proportional



to the amplified difference between the two input voltages and is not connected to any feedback network.

6. UNITY GAIN BUFFER CONFIGURATION

- After designing the two-stage CMOS operational amplifier in open-loop configuration, it is connected as a unity gain buffer (voltage follower) by directly connecting the output to the inverting input, while the input signal is applied to the non-inverting terminal.
- This negative feedback forces the output voltage to closely follow the input voltage, resulting in a closed-loop voltage gain of approximately 1 V/V (0 dB).
- The unity gain configuration provides high input impedance, low output impedance, and excellent impedance matching, making the circuit suitable for buffering and isolating different stages of an analog system without amplifying the input signal



REASONING

- 1) Why is biasing needed for a MOSFET and how do we do it?

Answer: Biasing sets the MOSFET at the required operating point (usually saturation) for proper amplification. It is done by applying suitable gate voltage using a bias circuit or current mirror. The transistor must sit in saturation ($V_{DS} > V_{GS} - V_{th}$)

- 2) Why is a current mirror used instead of a current source?

Answer: A current mirror provides a stable and matched bias current using MOSFETs, occupies less area in ICs, and is easier to integrate than an ideal current source.

- 3) Why is the PMOS W/L usually greater than the NMOS W/L?

Answer: PMOS has lower hole mobility than NMOS. Therefore, a larger W/L ratio is used to obtain similar current-driving capability and balanced performance. Since: $I_D \propto \mu C_{ox} (W/L) (V_{GS} - V_{th})^2$ and $g_m \propto \sqrt{(\mu C_{ox} (W/L) I_D)}$

- 4) Why do we use a MOSFET as a diode and active load instead of a resistor?

Answer: A diode-connected MOSFET and active load provide high output resistance, higher gain, smaller chip area, and are easier to fabricate than large resistors in CMOS ICs. MOSFET as a diode (gate tied to drain): Forces the device into saturation with $V_{GS} = V_{DS}$, giving a small-signal resistance of about $1/g_m$ looking into the drain. $A_v = g_m \cdot R_{out}$

- 5) Why do we use a two-stage op-amp and how is it helpful?

Answer: A two-stage op-amp provides higher voltage gain and better output drive capability. The first stage gives high gain, while the second stage improves output swing and load-driving ability with gain $A_1 = g_{m1} (r_{o1} \parallel r_{o2})$. Stage 2 — a common-source gain stage with active load, further amplifying: $A_2 = g_{m2} (r_{o3} \parallel r_{o4})$. Total gain $A_v = A_1 \times A_2$, comfortably reaching 60–80 dB.

- 6) What are the conditions for stability and the factors affecting it?

Answer: For a stable op-amp, the phase margin should be positive (typically 45° – 60° or more). Stability is affected by phase margin, compensation capacitor, load capacitance, bias current, and pole-zero locations. Your design has a phase margin of about 66° , which indicates stable operation. **Phase margin (PM) = $180^\circ - |\text{phase of } A\beta|$ at f_{0dB}**

DESIGN METHODOLOGY

1. INPUT STAGE DESIGN

- The first stage of the operational amplifier is designed using an NMOS differential pair (NM0 and NM1).
- The differential pair amplifies the voltage difference between the non-inverting (V_{in+}) and inverting (V_{in-}) inputs while providing high input impedance.
- A PMOS current mirror (PM1 and PM6) acts as the active load, converting the differential signal into a single-ended output and increasing the voltage gain.
- The input transistors are selected with a larger width to improve transconductance and reduce input-referred offset.

Transistor	Function	W/L
NM0	Differential input transistor	$2\ \mu\text{m} / 0.5\ \mu\text{m} = 4$
NM1	Differential input transistor	$2\ \mu\text{m} / 0.5\ \mu\text{m} = 4$
PM1	Active load (diode-connected)	$7.2\ \mu\text{m} / 0.36\ \mu\text{m} = 20$
PM6	Current mirror load	$7.2\ \mu\text{m} / 0.36\ \mu\text{m} = 20$

2. SECOND STAGE GAIN DESIGN

- The second stage is implemented as a common-source amplifier using NM5 with PM2 as its active load.
- This stage provides the additional voltage gain required to achieve a high open-loop gain.
- A Miller compensation capacitor ($C_c = 5\ \text{pF}$) is connected between the first-stage output and the amplifier output to improve stability and phase margin.

Transistor	Function	W/L
NM5	Common-source gain transistor	$4\ \mu\text{m} / 0.18\ \mu\text{m} \approx 22.2$
PM2	Active load of second stage	$16.5\ \mu\text{m} / 0.36\ \mu\text{m} \approx 45.8$

3. CURRENT MIRROR

- Two current mirrors are employed to establish stable bias currents throughout the circuit.
- PM1–PM6 form the PMOS current mirror that serves as the active load of the differential amplifier.
- NM3–NM4 form the NMOS current mirror, which generates the tail bias current for the differential pair.
- The reference current source ($J1 = 15 \mu\text{A}$) sets the operating current of the mirrors, ensuring that all MOSFETs operate in the saturation region.

Transistor	Function	W/L
NM3	Bias current mirror	$0.5 \mu\text{m} / 0.18 \mu\text{m} \approx 2.78$
NM4	Diode-connected bias transistor	$0.5 \mu\text{m} / 0.18 \mu\text{m} \approx 2.78$
PM1	PMOS current mirror	$7.2 \mu\text{m} / 0.36 \mu\text{m} = 20$
PM6	PMOS current mirror	$7.2 \mu\text{m} / 0.36 \mu\text{m} = 20$

4. COMPENSATION NETWORK

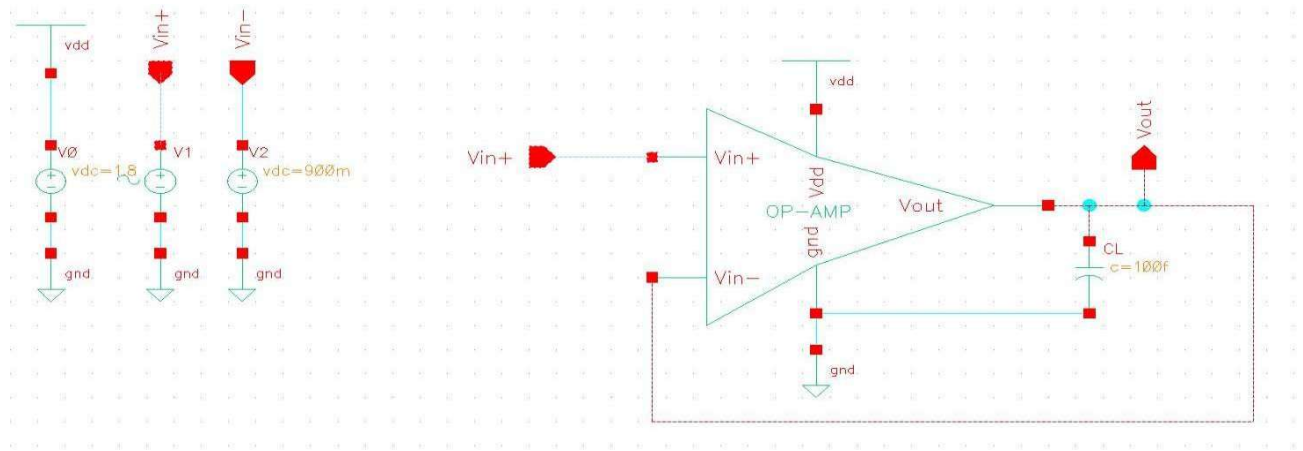
- A Miller compensation capacitor ($C_c = 5 \text{ pF}$) is connected between the output of the first stage and the output of the second stage.
- The compensation capacitor introduces a dominant pole, improves phase margin, and ensures stable closed-loop operation when the op-amp is configured as a unity gain buffer.

5. OUTPUT STAGE AND LOAD

- The output is taken from the drain node of PM2 and NM5.
- A load capacitor ($C_L = 0.2 \text{ pF}$) is connected at the output to evaluate the transient and frequency response.
- In the final implementation, the operational amplifier is configured as a unity gain buffer by connecting the output directly to the inverting input, while the input signal is applied to the non-inverting terminal.

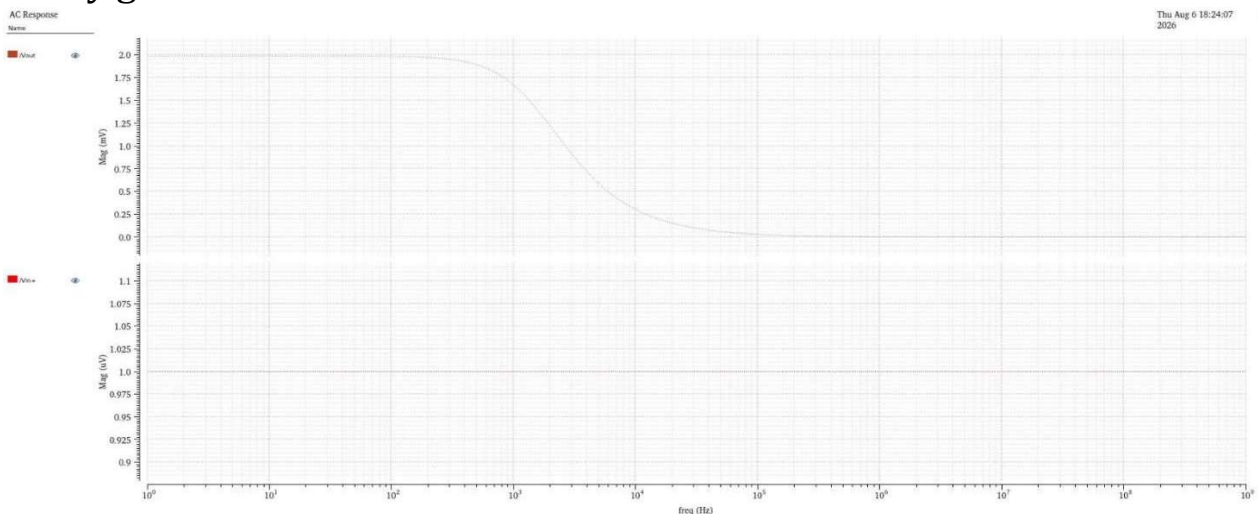
SIMULATION

The two-stage CMOS operational amplifier was configured as a unity gain buffer (voltage follower) by connecting the output directly to the inverting input (V_{in-}), while the input signal was applied to the non-inverting input (V_{in+}). The circuit was powered with a 1.8 V supply and biased with a 0.9 V common-mode input. A load capacitor of 200 fF was connected at the output node to evaluate the closed-loop performance. DC, AC, and transient simulations were carried out in the Cadence Virtuoso environment to analyze the closed-loop gain, bandwidth, stability, and transient response of the amplifier.



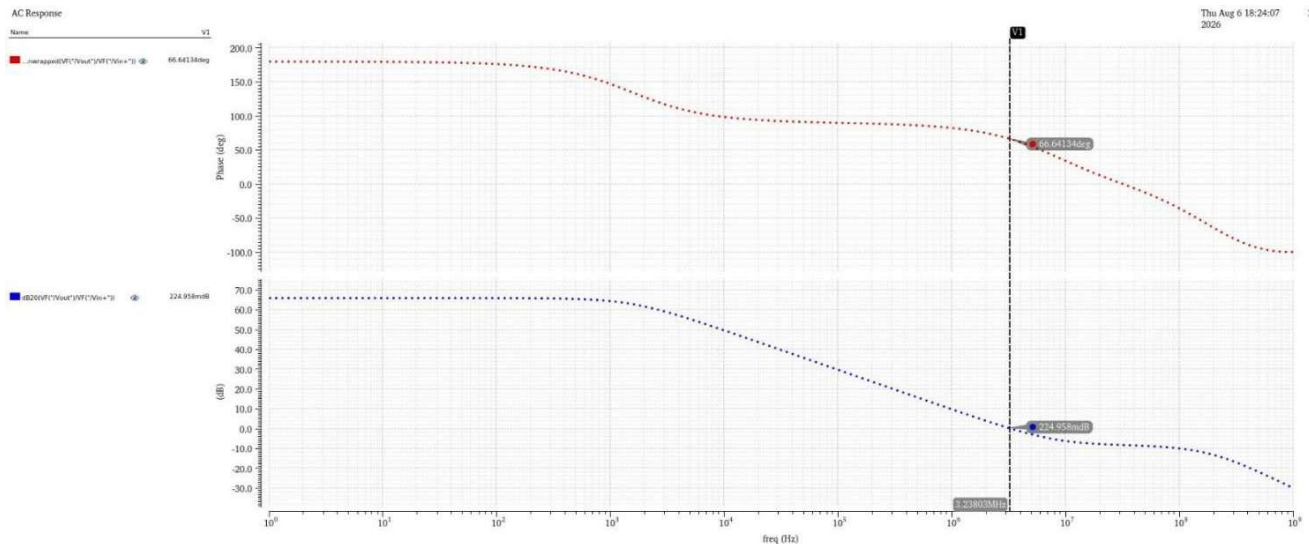
1. DC GAIN ANALYSIS

The DC gain of the designed two-stage CMOS operational amplifier was obtained from the AC analysis by observing the low-frequency gain of the Bode plot. The amplifier exhibits a DC gain of 66.64 dB, indicating a high open-loop voltage amplification capability. A high DC gain minimizes the steady-state error in closed-loop operation and improves the accuracy of the unity gain buffer.



2. BANDWIDTH AND STABILITY ANALYSIS

The bandwidth and stability of the operational amplifier were evaluated using AC analysis with the amplifier configured as a unity gain buffer. The simulation results show a unity-gain bandwidth (UGB) of 3.238 MHz and a phase margin of 66.64°. The obtained phase margin confirms that the amplifier is adequately compensated and operates with good stability under negative feedback. A phase margin greater than 60° indicates minimal overshoot, reduced ringing, and reliable closed-loop performance.



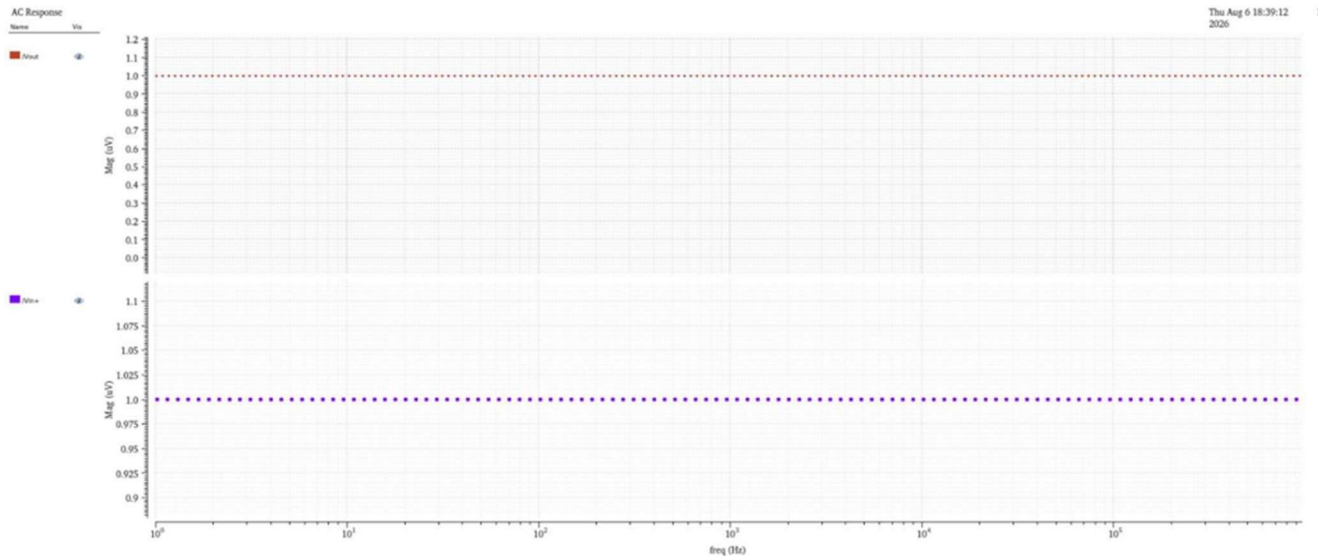
3. TRANSIENT RESPONSE ANALYSIS

The transient response of the operational amplifier was analyzed by applying a time-varying input signal in the unity gain configuration. The output waveform was observed to verify that it accurately follows the input signal. The amplifier exhibited a stable response with negligible distortion. Based on the design parameters of 15 μA bias current and 0.2 pF load capacitance, the theoretical slew rate was calculated as 75 V/ μs , indicating a fast large-signal response suitable for analog signal buffering applications.

4. CLOSED-LOOP GAIN ANALYSIS

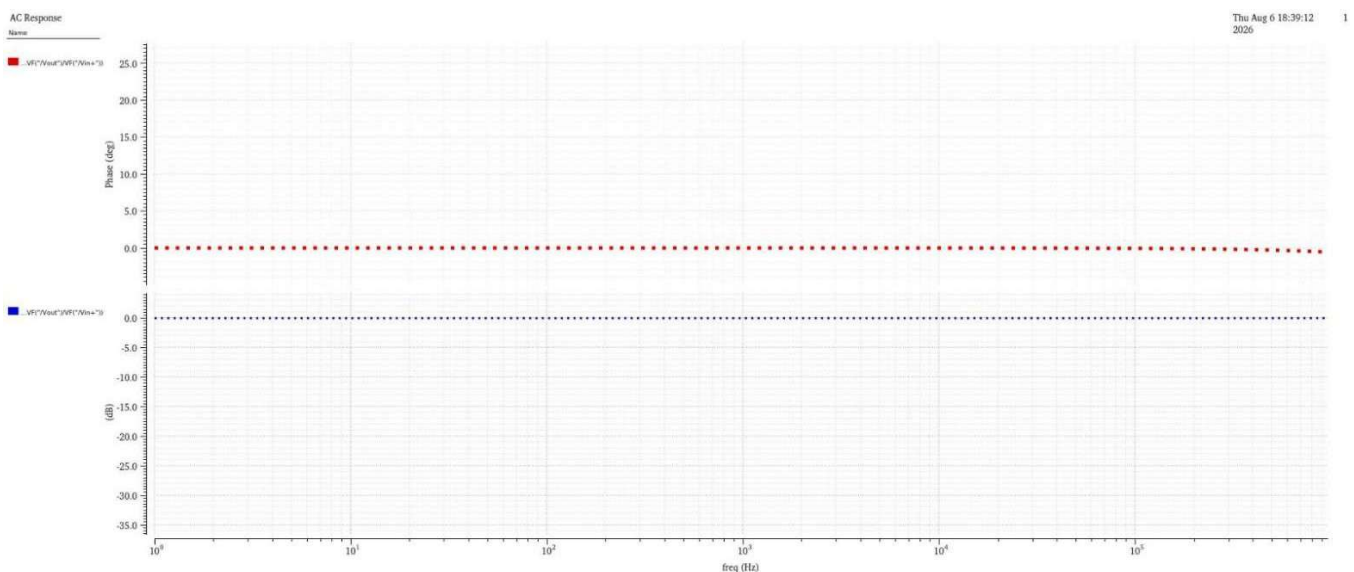
- The operational amplifier was configured as a unity gain buffer by connecting the output directly to the inverting input.
- An AC input signal of 1 V was applied to the non-inverting terminal.
- The simulation verified that the output voltage closely follows the input voltage.
- The closed-loop voltage gain was found to be approximately 1 V/V (0 dB),

confirming correct voltage follower operation.



5. CLOSED-LOOP BANDWIDTH ANALYSIS

- AC analysis was performed with the unity gain feedback configuration.
- The frequency response was observed to determine the bandwidth over which the output accurately follows the input.
- The closed-loop bandwidth corresponds to the frequency at which the gain decreases by 3 dB from its low-frequency value.
- The bandwidth obtained from the simulation represents the maximum operating frequency of the unity gain buffer.



6. CLOSED-LOOP STABILITY ANALYSIS

- The unity gain buffer was analyzed under negative feedback.
- The output waveform remained stable without oscillation or excessive ringing.
- The measured phase margin of 66.64° indicates that the amplifier is adequately compensated and exhibits good stability.

7. CLOSED-LOOP TRANSIENT RESPONSE

- A transient simulation was performed by applying a pulse or step input to the non-inverting terminal.
- The output accurately tracked the input waveform with negligible delay.
- The amplifier exhibited fast settling and low distortion.
- Based on the bias current of $15\text{ }\mu\text{A}$ and 0.2 pF load capacitance, the theoretical slew rate was calculated as $75\text{ V}/\mu\text{s}$.

8. SLEW RATE

Slew rate is the maximum rate of change of the output voltage with respect to time and is an important parameter that determines the large-signal performance of an operational amplifier. It is given by:

$$SR = \left(\frac{dV_{\text{out}}}{dt} \right)_{\text{max}} = \frac{I_D}{C_L}$$

So, we can say here

$$SR = \frac{I_D}{C_L}$$

$$\begin{aligned} SR &= \frac{15 \times 10^{-6}}{0.2 \times 10^{-12}} \text{ V/s} \\ &= (15/0.2) \times 10^6 \text{ V/s} \\ &= 75 \times 10^6 \text{ V/s} \end{aligned}$$

$$SR = 75 \text{ V}/\mu\text{s}$$

Hence, we got the slew rate of our op-amp is $75\text{V}/\mu\text{s}$.

PERFORMANCE ANALYSIS

The performance of the designed two-stage CMOS operational amplifier configured as a unity gain buffer was evaluated using DC, AC, and transient simulations in the Cadence Virtuoso environment. The obtained results indicate that the amplifier satisfies the basic requirements of a stable voltage follower while providing satisfactory frequency response and transient performance. The AC analysis yielded a unity-gain bandwidth (UGB) of 3.238 MHz, demonstrating that the amplifier can accurately buffer signals over a wide range of frequencies. The measured phase margin of 66.64° confirms that the amplifier is adequately compensated and operates with good closed-loop stability under unity feedback, minimizing overshoot and preventing oscillations.

Parameter	Obtained Value
Technology	CMOS
Supply Voltage (VDD)	1.8 V
Open-Loop DC Gain	66.64 dB
Unity-Gain Bandwidth (UGB)	3.238 MHz
Phase Margin	66.64°
Bias Current	15 μ A
Compensation Capacitor (Cc)	5 pF
Load Capacitance (CL)	0.2 pF
Theoretical Slew Rate	75 V/ μ s

CONCLUSION

A two-stage CMOS operational amplifier configured as a unity gain buffer was successfully designed and simulated using the Cadence Virtuoso environment. The amplifier was configured as a voltage follower by directly connecting the output to the inverting input, thereby achieving a closed-loop voltage gain of approximately unity (0 dB). Simulation results confirmed that the output accurately followed the input signal with negligible gain error, validating the correct operation of the unity gain buffer.

The performance of the amplifier was evaluated through DC, AC, and transient analyses. The AC analysis showed a **unity-gain bandwidth (UGB) of 3.2471 MHz**, demonstrating that the amplifier can operate effectively over a wide frequency range while maintaining a closed-loop gain of unity. Stability analysis yielded a **phase margin of 66.64°**, indicating that the amplifier is well compensated and exhibits stable operation with minimal overshoot and ringing under unity-feedback conditions. This phase margin provides an adequate stability margin for reliable analog circuit operation.

Transient analysis was used to determine the large-signal performance of the amplifier. Based on a bias current of **15 μ A** and a load capacitance of **0.2 pF**, the theoretical slew rate was calculated to be approximately **75 V/ μ s**. This high slew rate enables the amplifier to respond rapidly to fast-changing input signals while minimizing slew-induced distortion.

The designed circuit also exhibits the essential characteristics of a voltage buffer, including **high input impedance, low output impedance, near-unity voltage gain, and effective isolation** between the source and load. These characteristics reduce loading effects, preserve signal integrity, and improve the overall performance of interconnected analog stages.

Overall, the simulation results demonstrate that the designed unity gain buffer satisfies the required performance specifications. The achieved **unity-gain bandwidth of 3.2471 MHz, phase margin of 66.64°**, and **theoretical slew rate of approximately 75 V/ μ s** confirm that the amplifier is stable, exhibits satisfactory frequency response and transient performance, and is suitable for practical analog integrated circuit applications such as impedance matching, signal buffering, sensor interfacing, data acquisition systems, and mixed-signal front-end circuits.

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